Project Report: Creative mind Game

1. Project Overview:
   The project implements a number-guessing game where users try to guess a randomly generated number
   by the machine. The game includes user registration, customizable settings, and score calculation.
2. Features Implemented:

- Linked list for storing users and scores
- Random number generation with optional repeated digits
- Input validation for guesses
- Feedback for correct digits in correct/wrong positions
- History of guesses (optional)
- Scoring based on number complexity, attempts, and time

3. Technical Details:

- C++ standard libraries used: iostream, vector, string, algorithm, ctime, unordered_map
- Linked list implemented manually for storing user data
- Sorting algorithm used to display scores in descending order

4. Game Logic:

- Max 12 attempts allowed
- Time allocation per digit: 10 sec if digits unique, 7 sec if repeated
- Score formula:
  Score = length*10 + (repeatAllowed?5:0) + (historyShown?5:0) + max(0, 50 - tries*4 - totalTime/5)

5. Challenges Faced:

- Managing dynamic linked list for user data
- Calculating feedback accurately for digits in wrong positions
- Designing a simple menu system for multiple functionalities

6. Conclusion:
   The project demonstrates core concepts of C++ including data structures (linked list, vectors),
   random number generation, and basic I/O handling. The game is interactive and provides feedback
   based on user performance.